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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME1505-1 – INTEGRATED AUTOMOTIVE SAFETY
(Open Elective)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is a primary goal of consumer automotive safety tests?**
A) Improve vehicle aesthetics
B) Determine a vehicle's crashworthiness
C) Increase production speed
D) Assess fuel economy
2. **Which of the following is an environmental factor contributing to accidents?**
A) Weather conditions
B) Driver inattention
C) Vehicle malfunctions
D) Road sign visibility
3. **Which regulation body is responsible for establishing global safety standards?**
A) ISO
B) NCAP
C) SAE
D) NHTSA
4. **Which of the following is NOT a function of ADAS?**
A) Adaptive Cruise Control
B) Lane Departure Warning
C) Emergency Braking Assist
D) Tire pressure optimization
5. **Which sensor is essential for GPS navigation in ADAS?**
A) RADAR
B) GNSS
C) LIDAR
D) Ultrasonic
6. **What is the main advantage of LIDAR over RADAR?**
A) Better detection in foggy conditions
B) Higher resolution and accuracy
C) Longer detection range
D) Lower cost
7. **Driver drowsiness detection systems primarily monitor:**
A) Steering patterns
B) Vehicle speed
C) Road conditions
D) GPS signals
8. **What does "Sensor Fusion" refer to in ADAS?**
A) Combining multiple sensor data for improved accuracy
B) Using identical sensors for redundancy
C) Merging sensors with autonomous vehicles
D) Simplifying sensor designs for cost reduction
9. **Which system helps monitor road signs and alert the driver?**
A) Intelligent Headlight Control System
B) Traffic Sign Recognition System
C) Blind Spot Detection
D) Evasive Steering Support
10. **What is the primary benefit of 360-degree Surrounding View Systems?**
A) Enhancing long-distance detection
B) Monitoring vehicle speed
C) Assisting drivers during parking and manoeuvring
D) Providing weather condition alerts

11. **HIL in ADAS testing refers to:**
 A) Human-Interface Logic B) Hardware-in-the-Loop
 C) High-Intensity Light sensors D) Hazard Inspection Line
12. **Secondary restraint systems include which of the following?**
 A) Seatbelts B) Airbags
 C) Vehicle roll cages D) ABS
13. **Why are Anthropomorphic Test Devices crucial in crash testing?**
 A) They measure fuel consumption during crashes B) They simulate human injuries to assess safety
 C) They monitor vehicle performance D) They reduce crash simulation costs
14. **Which of the following best describes the role of biomechanics in crash testing?**
 A) Understanding human injury mechanics during collisions B) Analysing the aerodynamics of vehicle design
 C) Enhancing fuel efficiency D) Monitoring tire wear patterns
15. **Which of these materials is often used for crash energy absorption?**
 A) Carbon fibre B) High-strength steel
 C) Polystyrene D) Aluminium foam
16. **What does crash simulation testing aim to optimize?**
 A) Vehicle speed during impact B) Restraint system effectiveness
 C) Overall vehicle weight D) Driver reaction times
17. **Which test device is used to replicate human motion during a crash?**
 A) Gyroscope B) Anthropomorphic Test Device
 C) Inertial Measurement Unit D) Accelerometer
18. **Which NCAP program is specific to India?**
 A) ASEAN NCAP B) Bharat NCAP
 C) Euro NCAP D) Global NCAP
19. **What aspect does a 5-star NCAP rating typically indicate?**
 A) Superior crashworthiness and occupant protection B) Better fuel economy
 C) Enhanced vehicle performance D) Lower maintenance costs
20. **The primary goal of safety regulations is to:**
 A) Reduce vehicle manufacturing costs B) Standardize safety measures across manufacturers
 C) Increase vehicle fuel efficiency D) Encourage luxury features

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Analyze how regulations and safety ratings influence consumer behavior and vehicle design. Discuss the challenges in harmonizing safety regulations globally.	8	L*2	1	1
	b) Explain how machine learning and deep learning are utilized in ADAS. What is the significance of sensors and sensor fusion in this context?	8	L2	2	1
2.	a) Discuss how accident-avoidance strategies contribute to road safety. Compare traditional safety measures with modern AI-driven techniques.	8	L2	1	1
	b) Explain the working principle of Adaptive Cruise Control (ACC) with a block diagram. What are its limitations?	8	L1	2	2
3.	a) Explain how machine learning and deep learning technologies enable real-time decision-making for autonomous driving. Discuss the limitations of current AI models in safety-critical scenarios.	8	L1	2	1
	b) Define automotive safety and explain its importance in the context of modern road systems.	8	L1	1	1

Unit – II

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| 4. a) | Describe the mechanisms involved in primary restraint systems like seat belts. How do these mechanisms help in reducing the impact of a crash? | 8 | L2 | 3 | 1 |
| b) | Provide a historical background on the evolution of Anthropomorphic Test Devices. How have current ATDs improved biomechanics and injury assessments? | 8 | L1 | 4 | 1 |
| 5. a) | Draw a diagram showing the working of a crash simulation system and its use in testing restraint systems. Explain the steps involved and how computational modeling improves crash analysis. | 8 | L2 | 4 | 2 |
| b) | Explain the working of an active head restraint system and its importance in reducing whiplash injuries. | 8 | L1 | 3 | 1 |
| 6. a) | Provide a schematic diagram showing the working mechanism of an airbag system during a crash. Label and explain each step in the deployment process. Compare front and side airbags in terms of deployment mechanisms. | 8 | L2 | 3 | 2 |
| b) | Describe the various materials used in automotive safety, such as metals, resins, and fabrics. What are the requirements and applications of these materials? | 8 | L1 | 4 | 1 |

Unit – III

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| 7. a) | Draw and explain a flowchart of NCAP testing procedures. How do these tests influence public perception? | 8 | L2 | 5 | 1 |
| b) | Define ASIL (Automotive Safety Integrity Level) rankings and their importance in functional safety. | 8 | L1 | 5 | 1 |
| 8. a) | Explain the concept of functional safety in the context of ISO 26262. What are the steps involved in hazard analysis and risk assessment, and how are ASIL rankings determined? | 8 | L2 | 5 | 1 |
| b) | What are the major differences between NCAP regulations in India and global NCAP regulations? What improvements can be made to enhance vehicle safety in India? | 8 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
